■ Verdax

ML Pipeline Sustainability Impact Report

Company: MetaVerdax

Report Period: 2026-04-14 to 2026-04-14

Generated: 2026-04-14 09:16 UTC

Scope: 2026-04-14 to 2026-04-14 (inclusive, by date(timestamp))

Executive Summary

During the period 2026-04-14 to 2026-04-14, Verdax monitored **0 training pipeline runs** for MetaVerdax. Of these, **0 retrains were prevented** due to detected data anomalies, saving an estimated **38.6 kg of CO■ emissions** and **$120.00 in compute costs**.

Impact Metrics

All metrics below are date-scoped to 2026-04-14 through 2026-04-14 and derived from logged incidents/carbon records.

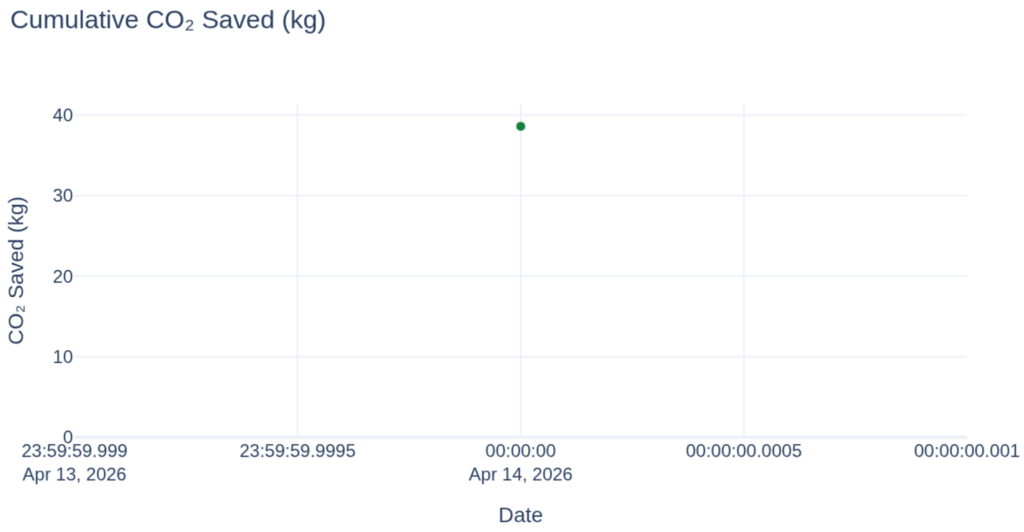
Metric	Value
Total Pipeline Runs	0
Retrains Prevented	0
CO■ Saved	38.6 kg
Energy Saved	100.0 kWh
Cost Saved	$120.00
Equivalent Flights Avoided	0.04
Equivalent Car km Avoided	184

Reporting Scope & Provenance

All totals are scoped by date(timestamp) using inclusive start/end filters. Carbon totals come only from prevented retrains recorded in the Verdax database ledger.

Incident rows in scope: 0; blocked rows: 0; carbon records: 1.

CO■ Saved Over Time



Incidents Over Time

No incidents recorded for this period.

Incident Log

No incidents recorded for this period.

Methodology & Sources

Verdax reports carbon impact using published training-energy benchmarks, regional grid-intensity factors, and explicit conversion formulas.

1. Carbon Calculation Framework

Formula: **Energy (kWh) × Grid Intensity (kg CO■/kWh) = CO■ (kg)**

Model size	Base energy (kWh)	Benchmark source
Small	10	Patterson et al. 2021 / conservative small-model benchmark
Medium	100	Patterson et al. 2021 / BERT-scale benchmark
Large	1,000	Patterson et al. 2021 / large-model benchmark
LLM	1,287,000	Luccioni et al. 2023 / BLOOM training emissions

2. Grid Intensity Factors

Region	kg CO■/kWh	Source	Last updated
us-east-1	0.386	EPA eGRID 2022	Apr 2026
eu-west-1	0.233	EPA eGRID 2022 / EU proxy	Apr 2026
ap-southeast-1	0.493	IEA regional average	Apr 2026
on_prem	0.475	Conservative global average	Apr 2026

3. Equivalency Conversions

1 transatlantic flight ≈ 1,000 kg CO■ (ICAO Carbon Calculator).

1 km driving ≈ 0.21 kg CO■ (EPA average passenger vehicle).

4. Limitations & Disclaimer

Estimates are order-of-magnitude benchmarks. Actual emissions vary by hardware, utilization, data-center PUE, and workload parallelism. Verdax uses conservative published averages rather than live cloud billing telemetry.

Methodology version: 1.0 — April 2026. Next review: October 2026.

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